Dhadheechi Ravva

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.85
Narayana Intermediate College, Pedda Ambarpet	Class XII (TSBIE)	2023	97.90%
Narayana Junior College, LB Nagar	Class X (SSC)	2021	100.00%

SKILLS

- **Programming Languages**: Python, C, Julia, SQL, familiar with C++, Go, Cuda, Javascript
- **Tools**: Git, Linux, Docker, LaTeX
- **Frameworks**: Numpy, Matplotlib, SciPy, PyTorch

RELEVANT COURSES

- **AI**: Foundations of Machine Learning, Deep Learning, Reinforcement Learning, Online Learning, NLP, CV
- **Computer Science**: Discrete Mathematics, Computer Architecture, Database Management Systems, Data Structures and Applications, Algorithms
- **Mathematics**: Matrix Theory, Probability and Random Variables, Calculus, Differential Equations, Applied Statistics

PROJECTS

Indian Handicrafts Portal: Full-Stack Cultural Heritage & AI Discovery Platform (Live / Link) (NLP Coursework/Group Project/ Prof. Maunendra Desarkar)

- **Developed** a static-first cultural discovery portal using **Next.js, React, TypeScript, and Tailwind CSS**, indexing **216+ GI-tagged Indian crafts** and deploying via Cloudflare Pages.
- **Architected** a hybrid search engine combining **TF-IDF retrieval** with **384-dimensional MiniLM semantic embeddings**, alongside an extractive text summarizer.
- **Integrated** interactive state-level visualizations and **AI4Bharat IndicTrans2** translation across 6 Indian languages, with client-side caching and duplicate detection.

LPCVC: Low-Power Video Action Recognition *(CV Coursework/Group Project/Prof. Konda Reddy) (Link)*

- **Developed** an end-to-end video action recognition pipeline using **PyTorch and R(2+1)D-18**, fine-tuned on Qualcomm's **QEVD dataset** across 92 exercise classes.
- **Engineered** multiprocessing video preprocessing with temporal sampling and **worker-stratified train/validation splits**, addressing class imbalance and data leakage.
- **Deployed** the model to the **Dragonwing IQ-9075 NPU** using Qualcomm AI Hub, establishing an **ONNX/TorchScript validation pipeline** for hardware inference.

RISC-V Simulator & Memory Hierarchy *(Coursework / Group Project / Prof. Rajesh Kedia)* **Jul-Nov 2024** (Link)

- **Built** a modular two-pass assembler and cycle-accurate **RISC-V (RV64I/RV32I) simulator in C**, supporting parsing, symbol resolution, machine-code generation, and function tracing.
- **Designed** an execution engine modeling registers, memory, control flow, and a **dynamic call stack**, with support for integer arithmetic and logical operations.
- **Implemented** a configurable **L1 cache simulator** supporting multiple associativity and replacement policies, with write-back/write-through models and hit/miss analysis.

Autoregressive Language Model (GPT-2) from Scratch (Link)

- **Built and pretrained a 124M-parameter GPT-2 decoder from scratch in PyTorch**, implementing **multi-head causal self-attention**, custom **LayerNorm**, **GELU** activations, and autoregressive generation pipelines with **temperature scaling** and **top-k sampling**.
- **Engineered downstream classification workflows** by mapping official pretrained weights, freezing early

transformer blocks, and fine-tuning the final layer and classification head, achieving **96.67% test accuracy** on spam detection.

EXPERIENCE

University of Tokyo -- Research Intern

Dec 2025 | *Tokyo, Japan*

- Studied and implemented recent methods in **3D computer vision and novel-view synthesis**, including **VGGT, 3D Gaussian Splatting, and DUSt3R**, gaining hands-on experience with multi-view geometry and neural scene representations.
- Reproduced research pipelines from academic papers and experimented with **camera pose estimation, dense 3D reconstruction, and scene representation**, developing practical familiarity with modern vision research workflows.

POSITIONS OF RESPONSIBILITY

- CR for the AI department during the academic year 2023-24
- SUGC (Senate undergraduate committee) for the entire UG'23 batch during the year 2023-24

EXTRACURRICULAR

- Played Chess as part of the National Sports Organization (NSO)
- Other sports: Tennis, Badminton, Volleyball, Cricket, Swimming
- Reading non-fiction books